

Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

Supplementary Material

Anonymous CVPR submission

Paper ID *****

This document holds the material referred to from the main paper. It covers the errors-in-variables treatment of the velocity regressor, the protocol behind the ablation measurement, the data paths of the DSEC measurements, additional implementation and control analyses, and, in Secs. 15 and 16, the predictor-side and label-side audits that main Sec. 4 summarizes. The RVT source revision is b80f5683a6e2d5de65d4bde8105d796ccb50dbb1, and the SSM-ViT source revision is 7c871b55a0c5f00673c2c3975f6b6a1c5adbab88. Section numbers below refer to this document. References labeled main Sec. n refer to the main paper. All Gen1 quantitative claims in both documents are validation-split claims obtained from ported released checkpoints, and sequence-bootstrap claims use the 406 sequences represented in the stored-detection evaluation.

1. The velocity regressor

Reproducibility. Code, configuration files, data-preparation scripts and the exact evaluation scripts used for every table and figure in the main paper and this supplement will be released upon acceptance. All reported numbers were produced with the released configuration.

The regression of Eq. 5 uses the finite-differenced label velocity $\tilde{v} = v + \eta$, where v is latent true velocity and η is label noise. Three error mechanisms enter, and they behave differently. The shared-noise term determines the estimator choice.

Classical dilution. Let the label center carry noise of standard deviation σ_c per frame and let the speed be formed by a central difference over $\pm\Delta$. The differenced speed then carries noise with variance

$$\sigma_\eta^2 = \frac{2\sigma_c^2}{(\Delta_f + \Delta_b)^2}, \quad (1)$$

and τ is attenuated toward zero by $\lambda_{\text{att}} = \text{Var}(v)/(\text{Var}(v) + \sigma_\eta^2)$. The noise is the label noise

of the dataset the fit runs on, so it is measured there. Sec. 3 solves for $\sigma_c = 0.6515 \text{ px}$ on Gen1 itself, and Eq. 1 is evaluated row by row instead of assuming a fixed cadence. On the reported rows $\text{Var}(\eta)$ is 1.0 against a $\text{Var}(\tilde{v})$ of 110.6 in px^2/s^2 , so $\lambda_{\text{att}} = 0.991$. Because the reported Eq. 2 also includes geometric controls and per-sequence effects, this marginal calculation is not the exact attenuation factor for the final multivariable coefficient. We use it only as a scale diagnostic showing that label-velocity noise is small relative to marginal velocity variation. Under the synthetic setup of Sec. 4, the centered estimator recovers 0.92 of the injected lag against the predicted 0.933.

Smoothing bias. On the uniform grid used for the label-noise calibration, a central difference returns the mean velocity over 2Δ instead of the instantaneous velocity. Expanding the trajectory about the query instant,

$$\frac{x(t + \Delta) - x(t - \Delta)}{2\Delta} = \dot{x}(t) + \frac{\Delta^2}{6} \ddot{x}(t) + O(\Delta^4), \quad (2)$$

so the regressor carries a systematic term of order $\Delta^2 \ddot{x}/6$. That is a bias in the regressor, not noise in it, and no errors-in-variables correction removes it. It is made small by making Δ small, which Eq. 1 says makes the dilution worse. The reported regression rows use the local secant $[x(t + \Delta_f) - x(t - \Delta_b)]/(\Delta_f + \Delta_b)$. Their median $|\Delta_f - \Delta_b|$ is zero, so the first-order acceleration term is absent in the typical row. The general expansion includes $(\Delta_f - \Delta_b)\ddot{x}/2$ and is covered by the stated local-secant interpretation.

The shared-noise term. The first two mechanisms are properties of the regressor alone. The third couples the regressor to the response. A forward difference $(c_{k+1}^* - c_k^*)/\Delta$ carries the label noise δ_k of the current center with weight $-1/\Delta$, and the residual $\hat{c} - c_k^*$ carries the same δ_k with weight -1 . The two therefore covary.

$$\text{Cov}(\hat{c} - c_k^*, v_f) = \frac{\text{Var}(\delta)}{\Delta} \quad (3)$$

That holds even at $\tau = 0$, and the induced slope is $\text{Var}(\delta)/\Delta \text{Var}(v)$. A backward difference carries δ_k with the opposite sign. The centered difference omits c_k^* and removes the direct same-frame shared-noise term. Residual bias from temporally correlated annotation errors is not identified by this construction, which is why the forward and backward fits are reported beside it as a control and calibrated in Sec. 4.

Instrumental-variable alternative. An instrumented estimate would replace the current speed by a velocity finite differenced from a strictly disjoint earlier label pair, on the argument that disjointness implies independent errors. DSEC provides a counterexample to assuming that disjoint label errors are generally independent. The third-difference kernel $(1, -3, 3, -1)$ forces $\rho(1) = -0.75$ for white noise, and the same code returns -0.740 on white noise and -0.3566 on the DSEC-Det labels. Pooled-ratio bias, heteroscedastic segments and heavy tails each returned the white-noise value through the identical pooling code, so the departure is a property of the labels. That measurement does not identify what the correlation structure is, beyond excluding a first-order autoregressive form. The paper does not report an instrumental estimate.

The construction term. A benchmark whose high-rate labels are constructed between anchors adds a term to $e_{\parallel}$, where $e_{\parallel} = \langle \hat{c} - c^*, \hat{u} \rangle$, that has nothing to do with the predictor. The chord of a curved image trajectory runs ahead of the truth throughout the interval and meets it at the anchors, which produces a signed, frame-locked, resetting residual for a predictor with no lag. Sec. 16.1 shows that exact adjacent-frame linear interpolation is inconsistent with the observed DSEC-Det label trajectories as the sole construction mechanism, and the Gen1 measurements of Sec. 15.4 are taken at label times.

2. The influence profile

Sec. 15.3 measures the window-ablation sensitivity profile of the released `rvt-t` checkpoint. The stored input is a stacked histogram of ten bins of 5 ms covering the 50 ms preceding the query time. For each sample the recurrent state is advanced over eight preceding steps so that the network is not read out of a zero state. The reference output is recorded, and then each bin in turn is zeroed and the output recorded again. A bin’s influence is the magnitude of the change its ablation produces in the emitted detection tensor, relative to its norm. That magnitude is averaged over 480 samples drawn from the first twelve lexicographically ordered released Gen1 validation sequences,

using the first forty post-warm-up samples from each sequence. This deterministic rule gives 480 samples. The measured ablation-sensitivity centroid is the weighted average $\sum_k s_k c_k / \sum_k s_k$ over the ten bin centers c_k , where s_k denotes the measured non-negative bin-ablation sensitivity. The ablated pass starts from the recurrent state *entering* the step, not from the state the reference pass returned. The two passes therefore share one history and differ only in the ablated bin. If the returned state is used instead, the passes are one step out of phase and the centroid moves by $+1.19$ ms.

For the recurrent-history measurement, the same deterministic sampling rule uses 576 samples, the first forty-eight post-warm-up samples from each of the same twelve validation sequences. The current window and the preceding nine windows form the 500 ms horizon, and the current window and preceding nineteen windows form the 1000 ms horizon. For each lag, the corresponding whole input window is zeroed while the saved reference history before that lag is restored, and the state is propagated through the intervening windows to the target step. The relative L_2 change in the target detection tensor is averaged over samples to give a non-negative window sensitivity q_{ℓ} . The horizon centroid is

$$c_{q,H} = \frac{\sum_{\ell=0}^{L_H-1} q_{\ell} [-(25 + 50\ell) \text{ ms}]}{\sum_{\ell=0}^{L_H-1} q_{\ell}},$$

where L_H is the number of windows in horizon H , and the newest-window share is $q_0 / \sum_{\ell} q_{\ell}$. The same saved reference history is used for the reference and ablated passes at each lag.

Two fills define q_{ℓ} , and the manuscript reports both. Under the *zero fill* the selected window is set to zero. Under the *swap fill* it is replaced by the window at the same within-chunk index drawn from an unrelated sample in a different validation sequence. That fill preserves a real window’s bin statistics and dynamic range while removing that window’s own content. Everything else is held identical across the two: the same lag, the same restored reference history, the same propagation through the intervening windows, and the same relative L_2 read-out. The two are interventions rather than ground truth and they do not agree on scale, which is why both are reported. From lag 0 to lag 19 the zero fill decays by a factor of 10.3 and the swap fill by 2.2. The ratio of swap to zero sensitivity rises from 2.43 at lag 0 to 11.47 at lag 19. Under the swap fill a window a second earlier still carries 46 % of the newest window’s sensitivity. Substantial long-lag sensitivity therefore persists under the on-manifold intervention, although its absolute magnitude remains instrument-dependent.

Two properties of that measurement bound it. Ablation reports the sensitivity of the output to a bin. The ablation profile is an operational input-sensitivity measure and is not, in general, a decomposition of temporal weights. For a nonlinear predictor it is a local instead of a global descrip-

tion. A single-bin ablation also leaves the remaining bins in place, so the measurement is a partial derivative of the output, not a decomposition of it.

Because the profile is defined by an intervention, the centroid is reported under four instruments instead of one. Zeroing a bin gives -23.81 ms, setting it to its dataset mean -26.35 ms, and replacing it with the same bin of an unrelated sample -25.01 ms. The gradient of the output norm with respect to the bin, which ablates nothing at all, gives -22.35 ms. The four span 4.01 ms. Two of them remove content, one substitutes foreign content and one perturbs nothing, so their agreement is not a property shared by construction. The property the manuscript uses the centroid for, that it sits tens of milliseconds before the label instant and well away from the fitted temporal coefficient, holds under all four.

The sampling rule is deterministic, not random, so the sequences are the resampling unit for its uncertainty. Resampling the 12 sequences with replacement gives a 95 % interval of $[-24.07, -23.56]$ ms for the zero-fill centroid, a standard error of 0.13 ms, and no interval wider than 0.51 ms under any of the four instruments. Dropping any one sequence moves the estimate by at most 0.08 ms, and the individual sequence centroids run from -24.63 to -23.20 ms. The choice of instrument therefore moves this centroid by 4.01 ms while the choice of sequences moves it by well under one. The instrument spread is the binding uncertainty here, and the sampling error is negligible beside it. Sequences outside the twelve are a separate question, answered only by measuring them.

Masking the leading bins of the stored input is not a route to the same quantity at a shorter support. For a fixed network of influence weights a_k on bin centers c_k , let a constant-velocity trajectory be $p(t) = p_0 + vt$. The conditional expectation after masking the leading j bins is $A_j p_0 + C_j v$ with $A_j = \sum_S a_k$ and $C_j = \sum_S a_k c_k$ over the retained set S . Reading a time off that requires the normalization C_j/A_j , which the network need not perform, and the result equals the retained-mass centroid only for non-negative weights.

3. The label-center noise of Gen1

The errors-in-variables treatment of Sec. 1 and the calibration of the velocity estimator both need the dispersion of the Gen1 label center about a smooth trajectory. The value solved for elsewhere in this work, 0.4974 px, is DSEC-Det's, and does not transfer to a different annotation effort on a different sensor, so it is solved again on Gen1.

For a track center series $x_i = f(t_i) + e_i$ on a uniform grid, the k -th difference has $\text{Var}(D_k) = \binom{2k}{k} \sigma^2$ plus a motion term of order k , so $\text{Var}(D_k)/\binom{2k}{k}$ falls toward σ^2 from above. Halving the sampling rate multiplies the motion term by 4^k and leaves σ^2 untouched, giving a sec-

ond equation in the same two unknowns, $\sigma^2 = (4^k V_{\text{full}} - V_{\text{half}})/(4^k - 1)$. Tracks are rebuilt by the same-class overlap linking the output-time measurement uses. That gives 691 tracks of at least twelve labels on uniform annotation grids with cadences of 250, 500, or 1000 ms, corresponding to 4, 2, or 1 Hz. The reported regression rows instead use the two adjacent label neighbors with variable intervals. Their row-wise noise variance is $2\sigma_c^2/(\Delta_f + \Delta_b)^2$, instead of a fixed-cadence approximation. Variances are trimmed at 5 %, because all four Gen1 box coordinates are stored as integers and a median absolute deviation on lattice-valued data locks onto discrete values.

The two-rate elimination agrees with the full-rate ladder to four decimals from $k = 5$ onward, which is the check that the motion term is negligible rather than assumed away. The ladder extrapolates to $\sigma = 0.6515$ px, of which 0.1443 px is the quantization of a 0.5 px center lattice, leaving 0.6353 px of annotation dispersion.

4. The sign-flip calibration

Equation 3 predicts an artifact whose size is set by the label noise and the observed velocity variance, so it is calibrated before the estimator is used. 2000 synthetic tracks of fourteen labels on the Gen1 grid are generated with quadratic motion. The synthetic experiment calibrates the shared-noise artifact and attenuation under quadratic motion. The simulated detector reports the true position at $t - \tau_{\text{true}}$ with its own independent error, and the label centers are corrupted at 0.6515 px.

With $\tau_{\text{true}} = 0$ the forward estimator returns -5.64 ms and the backward one $+5.84$ ms, summing to $+0.20$ ms, against the closed form -5.63 ms, the centered estimator returns $+0.00$ ms. Setting the label noise to zero returns all three to $+0.02$ ms, and the placebo stays at its null throughout, so it does not absorb the artifact and remains usable on real rows. In the synthetic physical-lag experiment, the centered estimator recovers 0.92 of the injected lag, consistent with the predicted attenuation of 0.933 for this noise.

5. The output-time specification sweeps

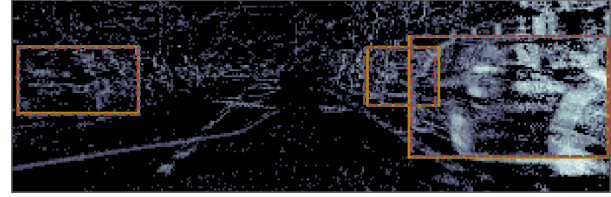
Lag recovery under the full control set. Adding $-\tau_{\text{inj}} \tilde{v}$ to the observed residual creates a known perturbation along the fitted temporal regressor. The coefficient is recovered at 1.000 of its injected value at 10, 25, 50 and 100 ms, showing that the geometric controls do not absorb this regressor. Injecting instead a purely geometric displacement of 0.05 px per px of box height moves the minimal specification's τ by $+0.63$ ms and the reported one's by less than 0.01 ms.

Sensitivity to axis pairing. The reported specification pairs box height and image row with $\hat{y}$ and box width and image column with $\hat{x}$. Entering all four on both axes gives

−3.62 ± 7.41 ms. Dropping each in turn moves τ by under 0.3 ms except for the image column, whose removal returns the placebo to its inflated value. Raising the overlap floor to 0.7 gives +0.49 ± 4.67 ms, the tightest of the sweep. Across every specification carrying geometric controls, including speed floors from 5 to 20 px/s, τ spans −20.2 to +11.1 ms, and none of them reaches the ablation-sensitivity centroid.

Association independent of nominal-time overlap. The rows are built by matching a detection to a label at the label’s nominal time. A gate on overlap could in principle discard exactly those detections whose temporal offset is large and select τ toward zero. Raising the floor cannot detect that, because it discards more. The informative direction is to open the gate or to remove it. `src/e62_association.py` rebuilds the rows from the five-checkpoint detection dump of Sec. 11 under 14 association rules and refits the reported specification under each. The rules are greedy and Hungarian assignment at overlap floors from 0.05 to 0.50. One rule never tests overlap at all, admitting a detection whose center falls within a fixed multiple of the label’s box scale. That dump is re-thresholded to the regression’s confidence floor of 0.1, but its non-maximum suppression was run at 0.65 against the 0.45 of the dump behind Tab. 8. Re-thresholding does not undo the looser suppression. The rebuild’s own 0.30 gate therefore keeps 22 518 rows against the 22 534 of the reported table and returns −2.96 ± 7.12 ms against −2.40 ± 7.18 ms, a difference of 0.08 standard errors. The sweep is read within the rebuild, as a comparison across association rules on one detection set. Across all 14 rules τ spans −8.96 to −0.90 ms, a spread of 8.06 ms against a median standard error of 7.12 ms. Every rule leaves τ within two standard errors of zero and leaves the ablation-sensitivity centroid more than two standard errors away. The overlap-free rule admits 22 895 rows against 22 518 for the rebuild’s own gate and gives −5.58 ± 7.86 ms against −2.96 ± 7.12 ms, a difference of 0.37 standard errors. Under the minimal specification the same sweep spans 12.01 to 16.35 ms, so neither specification’s conclusion is an artifact of the matching rule. Fig. 1 draws one frame of the sweep.

The bootstrap. Resampling sequences with replacement, 4000 times, gives a 95 % interval of [−17.0, +11.5] ms with 3 draws reaching the ablation-sensitivity centroid. The fit absorbs the per-sequence constant vectors by within-cell de-meaning, which reproduces the full design’s coefficient to 0.007 ms.



released detections
ground truth at the label instant
ground truth at the newest window's centroid, $\delta = -23.81$ ms

Figure 1. One validation frame of the sweep of Fig. 5b: released detections (solid), ground truth at its label instant (dashed) and displaced to the newest window’s ablation-sensitivity centroid (dotted). The frame is chosen by rule, the largest median label speed among frames with two labels above 20 px/s, its fastest label moves 45.2 px/s, so the two placements sit 1.08 px apart.

6. The per-object ratio

For comparison with the estimator in Eq. 5, we evaluate the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|\hat{v}\|$ on the same 22 534 rows. Across these rows, it has a sample standard deviation of 4170 ms and an interquartile range of 430 ms. Its mean over all rows is +24.22 ± 27.73 ms, close to the +23.81 ms coefficient corresponding to the newest-window ablation-sensitivity centroid under the convention of Eq. 5. Above a 15 px/s floor it gives +13.34 ± 6.17 ms. Because the ratio has no intercept, a velocity-independent spatial bias can appear as an apparent temporal offset after division by speed. Eq. 5 includes an intercept.

7. The port of the released checkpoint

The released `rvt-t` checkpoint loads into a model built from the repository’s own configuration files with 0 missing and 0 unexpected keys at 4.41 M parameters. It runs forward under a current framework without the training code it was serialized from. Both predictor measurements run on that port. A key count is a structural check, so two numerical ones are reported alongside it.

Parameter counts against the published ones. Summing the tensors each checkpoint carries gives 4.41, 9.87 and 18.54 M for the three RVT releases and 9.68 and 18.19 M for the two SSM-ViT releases. The largest deviation from the counts printed in the two papers is 0.051 M, so the ported models are the architectures the releases describe rather than approximations of them.

Independent evaluator cross-check. Every level in this paper comes from its own evaluator, so each detection dump is rescored with `pycocotools` at AP@[.50:.95] on the identical arrays. For `rvt-t` the two give 38.95 and 38.90 points, and across all five checkpoints they disagree by at most 0.101 points. The reported levels are therefore a property of the detections, not of the scorer.

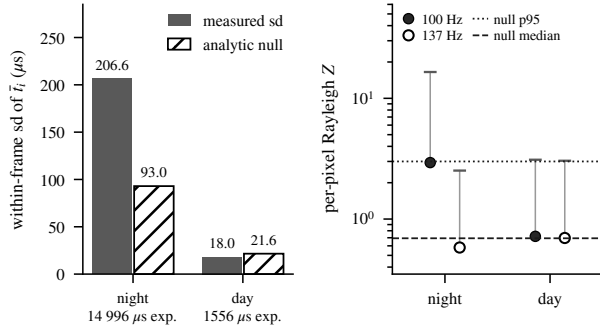


Figure 2. Left: within-frame dispersion of $\bar{i}_t$ against its analytic null in the two exposure strata, 1134 frames of `zurich.city_09.a` at 14 996 μs and 20 frames of `interlaken_00.c` at 1556 μs . The daytime frames are selected by retaining frames with at least three qualifying boxes. Right: per-pixel Rayleigh statistic $Z = R^2/N$ at 100 Hz and at 137 Hz for the same two sequences, median with a whisker to the ninety-fifth percentile, against the $\text{Exp}(1)$ null. Both contrasts run in the same direction, and Sec. 16.4 reports the controls applied to that dispersion.

Arm	frames	sd	null	excess
Ceiling sequence, all pixels	1134	206.6	93.0	183.6
Phase-locked pixels removed	1134	205.1	97.1	180.6
Random pixels, matched size	1133	206.3	94.8	182.9
Random-position boxes	1123	338.2	134.5	306.3
<i>by each box's own 100 Hz modulation depth m</i>				
$m \in [0.122, 0.410)$	270	119.3	74.6	93.1
$m \in [0.410, 0.467)$	243	149.1	76.2	128.1
$m \in [0.467, 0.534)$	207	156.6	85.1	131.5
$m \in [0.534, 1.050]$	265	227.2	91.5	208.0

Table 1. Per-object event centroid inside one published exposure window of `zurich.city_09.a`, in μs , each entry the median over frames of a within-frame statistic against that frame's own null $w^2/12N_i$. The first block's excess follows the Sec. 16.4 statistic summarized over frames. Rows two and three remove the 14 334 pixels phase-locked at 100 Hz and a random subset of that size, row four places boxes of the same shapes at random positions. In the lower block a frame enters a stratum when 3 of its boxes fall in it, so those frame counts do not partition the first row's.

Validation-split and checkpoint-port limitations. The published Gen1 tables are reported on the *test* split. The release distribution available to this work carries the validation split alone, 429 sequences with no test directory. The levels above are therefore validation-split levels, and the published numbers are used only for the parameter comparison and for the ordering within RVT, which is preserved. A pass of the original training-code path on identical samples, to give a per-sample output difference against the port, requires the training framework the checkpoints were serialized from. That pass is left to the artifact release. Of the 429 released validation sequences, 406 enter the stored-detection evaluation used for sequence resampling. The remaining 23 are excluded because they contribute no stored-detection rows. All bootstrap results in Secs. 12–13 resample these 406 sequences.

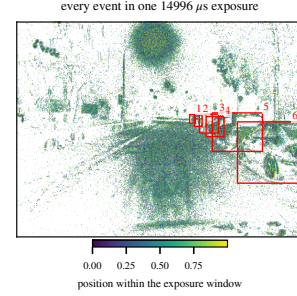


Figure 3. Every event inside one published 14 996 μs exposure window of `zurich.city_09.a`, colored by its position within that window, with that frame's released boxes numbered left to right. The frame is the median-dispersion one of 602 eligible candidate frames with at least six qualifying boxes, not the largest.

Sequence	boxes	m	Z_{100}	Z_{137}	locked
<code>zurich.city_00.a</code>	2914	0.428	117.6	3.15	100.00 %
<code>zurich.city_01.a</code>	2134	0.440	99.1	4.69	99.63 %
<code>zurich.city_02.a</code>	619	0.445	68.4	2.75	99.68 %
<code>zurich.city_03.a</code>	241	0.489	310.9	25.85	95.44 %
<code>zurich.city_09.a</code>	6997	0.466	251.5	7.82	99.23 %
<code>zurich.city_10.a</code>	3611	0.438	111.6	3.33	100.00 %

Table 2. Every DSEC training sequence whose exposure is pinned at 14 996 μs , measured per labeled box over the events inside that box during that frame's own published exposure window, with at least 400 events per box. m is the median over boxes of the modulation depth $m = 2|C|$ at 100 Hz. Z_{100} and Z_{137} are the median over boxes of the Rayleigh statistic at 100 Hz and at the off-frequency control on the same events. The analytic $\text{Exp}(1)$ median of 0.693 does not describe either column, which is why the two are compared with each other. The locked fraction denotes the fraction of boxes whose local excess at 100 Hz exceeds the ninety-ninth percentile of the same quantity at 137 Hz.

8. The DSEC measurements

Data path. For each labeled frame of a sequence, the frame's own exposure interval $[a, b]$ is read from the dataset's published `image_exposure_timestamps_left` file. The event stream is sliced to that interval through the release's `ms_to_idx` index, and the events inside each released box are collected. The `ms_to_idx` index resolves to whole milliseconds, so the slice it returns is filtered to $a \leq t < b$ in microseconds before any statistic is formed, and the analytic null is computed for the same interval. Events and frames share one clock in this dataset, so no calibration step enters. For the dispersion of Sec. 16.4 a box contributes only if it carries at least 200 events in the window, and a frame only if at least 3 boxes qualify. The reported statistics are medians over frames of a within-frame quantity, so no frame is weighted by its object count. For the per-box statistics of Table 2 a box contributes with at least 400 events and no frame-level requirement applies. The selected qualitative frame is shown in Supplement Fig. 3, and

the paired daytime control is shown in Supplement Fig. 2.

For frame f , let I_f be the qualifying boxes, let N_{fi} be the event count inside box i , and let $\bar{t}_{fi} = N_{fi}^{-1} \sum_{k \in I_f} t_k$ be its event-time centroid relative to the frame's mid-exposure. The observed within-frame dispersion and its frame-specific analytic null are

$$d_f = \left[\frac{1}{|I_f| - 1} \sum_{i \in I_f} (\bar{t}_{fi} - \bar{t}_f)^2 \right]^{1/2},$$

$$n_f = \left[\frac{1}{|I_f|} \sum_{i \in I_f} \frac{w_f^2}{12N_{fi}} \right]^{1/2}.$$

where $\bar{t}_f = |I_f|^{-1} \sum_{i \in I_f} \bar{t}_{fi}$ and $w_f = b_f - a_f$. The frame-level excess is $e_f = [\max(d_f^2 - n_f^2, 0)]^{1/2}$, and the reported entries are the medians of d_f , n_f , and e_f over eligible frames.

The stereo exposures. The left and right exposures open at the same microsecond in 100 % of the 7080 frames checked and differ only in when they close. The published image timestamp equals the mean of the two mid-exposures to within 1 μ s.

Archive version and track association. DSEC-Det ships more than one annotation version. The numbers in Sec. 16.1 describe the archive `dsec-det-left-object-detections.zip`, 4655023 bytes, retrieved 2026-09-01. Whether those labels are tracker output on the RGB frames, which the dataset's own description states, is not distinguishable from human annotation by a second-difference residual. On the Gen1 side the `track_id` field is a per-frame index instead of a persistent identifier. Association for the output-time fit is rebuilt by same-class overlap between adjacent label times, at which 79.0 % of boxes link.

Null. For events uniform on $[a, b]$ the mean of N_i times has variance $w^2/12N_i$ with $w = b - a$. Each frame therefore carries its own null, formed from the published width and the observed counts. The excess of Sec. 16.4 is summarized over frames rather than formed from the medians of the two summaries. That is why the excess column of Table 1 is not $(\text{sd}^2 - \text{null}^2)^{1/2}$ of the two columns beside it.

Phase test. The per-pixel test at frequency f takes the event phases $\phi_k = 2\pi f t_k \bmod 2\pi$ of a pixel, forms the resultant length R and reports $Z = R^2/N$, which is $\text{Exp}(1)$ under uniform phase. Raw modulation depth is not usable at

these counts, because for N events at random phase the resultant is already of the size of the effect. Pixels enter with at least 100 events over the tested slice, and every fourth event is used to reduce sample size. The off-frequency and daytime controls remain close to their nominal reference under the same procedure. Two arms hold the estimator to its null, an off-frequency arm on the same events and the same frequency on daytime events. The fraction of locked pixels is reported relative to live pixels in Sec. 16.2 and separately relative to the full sensor. The two denominators differ, and each statement identifies its denominator.

The per-box statistics of Table 2. The same estimator is applied to the events inside one released box during one published exposure window rather than to the events of one pixel over a long slice. For box i the depth is $m = 2|C|$ and the statistic is $Z = N_i|C|^2$ with $\omega = 2\pi f$ and $C = N_i^{-1} \sum_{k \in i} e^{-i\omega t_k}$, phases referred to that frame's mid-exposure instant. A finite exposure does not produce uniform phase at a tested frequency, so the pixel-level $\text{Exp}(1)$ null is not assumed for these per-box values. The depth is reported descriptively, without a random-phase floor. Boxes are matched to exposure windows by nearest mid-exposure instant within half a frame period. The off-frequency arm at 137 Hz is run on exactly the same events. The per-box 137-Hz arm is used as a same-event descriptive control, not as an independent spectral-resolution test. The illumination attribution is supported primarily by the long-slice 100-Hz harmonic structure and the daytime control.

Pixel exclusion and modulation strata. The exclusion mask of Table 1 is the set of pixels reaching $p < 10^{-3}$ in the per-pixel test. The matched control draws a random pixel set of exactly the same size, so the two arms remove the same number of pixels. The mask is built from a six-second slice at every fourth event. The stratification uses no mask. The quartiles of m are taken over all 6841 qualifying boxes, and a frame contributes to a stratum when at least 3 of its boxes fall inside it. The frame counts of that block are therefore not a partition of the first row's.

9. The permutation null

The analytic null $w^2/12N_i$ for the Sec. 16.4 statistic is the variance of a mean of N_i draws uniform on the exposure. It assumes the events of a box arrive independently and uniformly.

The permutation null does not assume independent or uniform event arrivals. Instead, it conditions on the pooled frame-level event times and each box's event count while randomizing box assignment. For each frame the events of all qualifying boxes are pooled and reassigned to boxes at random, preserving each box's count. The exposure, the

flicker and the scene’s burstiness all survive that reassignment, only which box an event belongs to is destroyed. Over 1134 frames and 200 draws each it returns $78.6 \mu\text{s}$, 0.85 times the analytic null rather than above it, and 94.1 % of frames exceed their own permutation null. A burst common to the frame displaces every $\bar{t}_i$ in the same direction and cancels in their dispersion, which is why the statistic was already insensitive to it. For a separate random-position draw evaluated against the permutation null, boxes of the same shapes give $339.4 \mu\text{s}$ against $113.3 \mu\text{s}$. Thus the dispersion is not specific to annotated object regions, although object regions may still affect it.

10. The carried state of the SSM release

Sec. 15.3 measures five released checkpoints of two architectures. The second is SSM-ViT (Zubic et al., CVPR 2024), a fork of the RVT repository in which each stage’s ConvLSTM is replaced by an S5 layer. It distributes RVT’s own preprocessed Gen1 and builds its window boundaries with the same statement of the same script, so the window convention of Sec. 15.2 holds there unchanged.

Its recurrent state does not. Let s denote the carried state entering the first step of a chunk. That release injects it as $\bar{\Lambda}_0 \leftarrow \bar{\Lambda}_0 \odot s$ before an associative scan whose binary operator is $(a_i, b_i) \circ (a_j, b_j) = (a_j a_i, a_j b_i + b_j)$. The scan’s outputs are the b components, and b_p is built from $a_1, \dots, a_p$ and $b_0, \dots, b_p$, $\bar{\Lambda}_0$ enters only the a component of the first element and therefore no output. The carried state is inert.

We verify this code-level property empirically. Driving each model the way its own release drives it and zeroing the state entering a chunk of 8 windows, the relative L_2 change in the emitted detection tensor is 0.00000 at every position for SSM-ViT. For `rvt-t` it is 0.0616 at position one, falling to 0.0304 at the last position. A model that ignored its history altogether would return zero to both arms, so the control ablates the window one position earlier inside the same chunk. SSM-ViT answers 0.0276 there, falling to 0.0106. It is recurrent within a chunk, and only across chunks is the state lost.

11. Effect of temporal displacement on checkpoint ordering

Sec. 15.5 reports that displacing the ground truth to the newest window’s ablation-sensitivity centroid costs 0.04 points on one checkpoint. Whether the interval changes a *comparison* needs more than one detector, so all 5 released checkpoints of Sec. 15.3 were run over the Gen1 validation split under one protocol. That protocol fixes the same frames, the same confidence floor of 0.01 and non-maximum-suppression overlap of 0.65. The ground truth is identical across the five and identical to the dump Sec. 15.5 already uses. Each was then scored at the label instant and

again at the displacement it prefers, which ranges from -15 to 0 ms across the five.

stratum	ordering at the label instant
all moving	$b > s > t > S5-B > S5-S$
10–25 px/s	$S5-B > b > t > S5-S > s$
25–50 px/s	$S5-B > b > S5-S > s > t$
≥ 50 px/s	$b > t > S5-S > s > S5-B$

Table 3. Ordering of the 5 released checkpoints at the label instant, by object speed, t , s , b are the RVT capacities and S5-S, S5-B the SSM ones. Scoring each at its own preferred displacement leaves every one of these orderings unchanged. A common displacement sweep reveals one localized inversion between `rvt-b` and S5-S in the 25–50 px/s stratum. Ranking varies across speed strata, and the common sweep produces no other pairwise inversion. `s5vit-base` is first in the middle two strata and last in the fastest, and `rvt-t` is third overall, last between 25 and 50 px/s and second above 50.

The own-preferred-offset scores preserve the validation-split ordering in each speed stratum (Tab. 3). Under a common displacement sweep, the 25–50 px/s stratum contains a localized ranking inversion between `rvt-b` and S5-S. The other evaluated strata contain no sign-changing pair. The aggregate ranking and the fast-object ranking have different leaders, so the strata are reported separately.

The released streaming evaluation steps through the split in non-overlapping chunks of 21 windows (sequence.length in config/dataset/gen1.yaml, start.indices in sequence_for_streaming.py steps by exactly that). The available recurrent history of a released detection therefore depends on where its frame falls inside that chunk, from a full 21 windows at the last position to the current window alone at the first. This is reported as a property of the release that its documentation does not state, which is the same class of undeclared quantity the rest of this paper measures. It is not reported as a claim about what the architecture could support if the state were carried.

12. Timing differences against the sampling uncertainty of the score

Sec. 15.5 reports that displacing the ground truth across the measured interval moves mAP by 0.04 points and reorders no checkpoint. On a fixed split both are deterministic functions of the stored detections, so neither is interpretable without the uncertainty that attaches to reading them as statements about the sequence population the split samples. Frames inside a sequence are dependent, so the re-sampling unit is the sequence. A cluster bootstrap over the 406 validation sequences was run for all 5 checkpoints at each displacement, resampling sequences with replacement and rescored from the stored detections.

Two standard errors must be distinguished. A claim

made *within* one paper compares that paper’s own score at two displacements on one set of detections. The relevant quantity is the paired $\text{SE}(\Delta\text{mAP})$, which is 0.035–0.058 points across the five checkpoints. A claim that compares *across* papers compares two absolute numbers, and the relevant quantity is $\text{SE}(\text{mAP})$, which is 2.19–2.82 points, fifty times larger.

Two quantities that are easy to conflate are therefore reported separately, because they are referred to different baselines. The first is the displacement *from the maximum of the sweep* at which the loss reaches two standard errors: 13.7–19.8 ms paired and 153–174 ms in the single-score sense. The second is the effect of the measured interval itself, referred to zero displacement, which is 0.041–0.081 points across the 5 checkpoints against a paired two-standard-error band of 0.070–0.116 points. The second clears its band for 1 of 5 checkpoints. The measured interval therefore sits at the paired sampling-resolution floor under sequence resampling. The identifiability conclusion rests on where the evidence lies, not on that comparison. The 0.04-point figure measures the sensitivity of the pooled benchmark to the interval, which follows from the benchmark’s speed composition (Tab. 9) and places no bound on the interval itself.

The same computation gives the timing difference a *leaderboard* could resolve. Let D^* be the smallest common displacement at which an adjacent pair in the validation-split ordering exchanges places. Over all moving boxes $D^* = 43.7$ ms, $1.83\times$ the measured interval, and in the 25–50 px/s stratum $D^* = 11.2$ ms, or $0.47\times$. A benchmark whose ordering only responds to timing displacements of this size cannot be used to certify that a timing convention is immaterial.

That ordering is itself unstable. The reference ordering here is the one the 5 checkpoints take on this validation split at the nominal label time. It is not the ordering of the published Gen1 tables, which are reported on the test split this release does not carry (Sec. 7). Resampling the same 406 sequences and re-ranking within each replicate, that validation-split ordering is reproduced in 80.5 % of replicates on the moving subset and 42.5 % over every labeled box, the population the released evaluation scores. Its tightest adjacent pair is separated by 0.050 points against a paired standard error of 0.53 and exchanges places in a fraction 0.48 of replicates. Two readings are consistent with the same evidence: that the correction changes no ranking on this split, and that the ranking is unstable under sequence resampling. The measured sampling uncertainty separates them in favor of the second.

Normalized image speed and timing resolution. The natural reading of Tab. 9 is that a benchmark of faster objects would resolve the interval. What an IoU-thresholded

metric registers is displacement relative to box size, so the scale-free quantity is $r = |v|/\sqrt{wh}$, box-widths of motion per second, which is invariant to sensor resolution and therefore comparable across benchmarks. A single fixed summary of r , such as its median, does not explain the observed resolution across strata. Scaling δ by each stratum’s median r increases the dispersion of the 5 %-loss timescale from 0.20 to 0.60, and no fixed quantile improves on the unscaled result.

mAP thresholds each box instead of averaging over them. Under the single-threshold caricature, in which a box loses its match once displaced past d^* box-widths and is untouched before, the fraction of boxes lost at displacement δ is $P(r > d^*/\delta)$. The displacement at which a loss level p is reached is $D(p) = d^*/Q_{1-p}(r)$. The scale-setter is the quantile *matched to the loss level being read off*, not the median or any fixed quantile. Scaled that way the strata do collapse, to a coefficient of variation of 0.042 at the 5 % level over three strata and 5 checkpoints, with $d^* = 0.0628 \pm 0.0029$ box-widths. That constant also follows from the matching geometry. Displacing a square box by a fraction d of its width gives $\text{IoU} = (1 - d)/(1 + d)$, so d^* is the displacement that breaks a match at $\text{IoU} = 0.882$. That threshold lies inside the 0.50:0.05:0.95 range this mAP averages over and near its top, where the first few percent of loss arise. The model degrades as the caricature should, to 0.10 at 10 % loss and 0.28 at 25 %. The ≥ 50 px/s stratum is the outlier at every loss level, with d^* approximately $1.8\times$ larger than the common value. Its lower mAP(0) may leave less high-IoU headroom for displacement, but no downstream claim depends on this interpretation.

Under this caricature, timing resolution is governed by the upper tail of r . Stratification increases the relevant quantile but also increases the standard error of ΔmAP as $1/\sqrt{n}$. The 25–50 px/s and ≥ 50 px/s strata yield net resolution factors of $0.42\times$ and $0.27\times$, respectively. Neither resolves the interval better than the full moving-box population.

On the DSEC-Det labels, associated by the `track_id` the release ships, not by IoU linking, r has median 1.31 s^{-1} against Gen1’s 0.095 s^{-1} . On the quantity that actually governs resolution the gap is smaller: $Q_{95}(r) = 5.59$, compared with 0.70 on Gen1, a $7.97\times$ difference. This label-composition contrast shows that DSEC-Det has a substantially heavier upper tail of normalized image speed than Gen1. It supports the narrower conclusion that Gen1’s timing insensitivity should not be generalized to event-based detection benchmarks with different motion and box-size distributions.

13. Recurrent support under the released streaming evaluation

The previous section established that the carried state of the SSM release is inert. Under its own streaming evalu-

ation the recurrent history behind a detection is bounded by the current chunk of 21 windows. That bound is not constant across detections: a frame at the first position of a chunk carries one window, and a frame at the last carries 21. The reported Gen1 number averages over all 21 positions, and neither the metric, the protocol, nor the published table names the variable.

The position is recoverable without running anything. The release derives its chunk starts from the index files it ships. The first chunk begins at

```
max(objframe_idx_2_repr_idx[0]
    - sequence_length + 1, 0)
```

and later chunks step by `sequence_length`. The position of every labeled frame therefore follows from those files alone. Reconstructing it over the 20 296 labeled frames reproduces, position by position, the positions the instrumented dumper records at run time.

Comparing the frames at positions 0–3 with those at positions 16–20 confounds support with scene content, since these are different frames. The three RVT checkpoints control for that. They see the identical frames in the identical order, but their ConvLSTM state is carried across chunks. Position in the chunk is not a support variable for them, and any difference they show is scene content. Differencing removes it. Sec. 13.3 measures that insensitivity on the same checkpoints instead of assuming it. Tab. 4 scores every labeled box, the population the released numbers use, and bootstraps over the 406 sequences.

checkpoint	short (1–4 win)	full (17–21 win)	raw Δ mAP (pt)	net of RVT control (pt)
rvt-t	38.180	38.194	0.01	-0.67 ± 0.43
rvt-s	38.371	39.167	0.80	0.51 ± 0.48
rvt-b	41.561	42.126	0.56	0.16 ± 0.55
S5-S	31.578	36.410	4.83	4.37 ± 0.61
S5-B	34.445	40.226	5.78	5.32 ± 0.53

Table 4. mAP over every labeled box, by the amount of recurrent history the released streaming evaluation gives the frame. Raw Δ mAP is the full column minus the short one, the quantity the top panel of main Fig. 1 plots, so positive means longer history scored higher. The last column subtracts the placebo trend from it, using the mean of the other two RVT rows for an RVT row and the mean of all three for an SSM row. The RVT rows are the placebos: their state crosses chunk boundaries, so position carries no support for them. Net of that control the SSM rows gain 4.37 and 5.32 points ($z = 7.2$ and 10.0), against at most 0.67 points in magnitude ($z = 1.56$) for any placebo. The short and full columns carry three decimals, so differencing them reproduces the raw column exactly. Cluster bootstrap over 406 sequences.

The RVT placebos isolate chunk-position effects under the parallel-trends assumption, which Sec. 13.3 tests directly by shifting the boundary under the RVT checkpoints themselves. The RVT checkpoints span the same 38.180–41.561 mAP range and are scored on the same

frames. Their rows show no comparable shift, which argues against frame difficulty at early chunk positions as the source (Fig. 4).

A permutation null tests whether the reconstructed correspondence itself can account for the effect. Chunk position is reshuffled among the frames of each sequence. That preserves the scene, the sequence lengths, the label density and the sizes of both blocks, and destroys only the correspondence between a frame and the history the protocol gave it. Over 200 reshuffles the difference-in-differences is centered on zero with a standard deviation of 0.55 points. The observed values lie 9.8 and 7.6 standard deviations outside it ($p = 0.005$, the smallest value 200 reshuffles can return), while every placebo stays within 1.52. The effect disappears when the correspondence between a frame and its reconstructed history is permuted.

A second instrument uses neither the ground truth nor the evaluator. Mean detection confidence is a label-free statistic, and it is flat in chunk position when the available support does not vary with that position. For S5-S it rises monotonically from 0.463 at one window to 0.574 at seventeen, 23.8 %, while `rvt-t` stays flat across the same positions. The shift therefore appears upstream of the matching rule.

The comparison can also be made against the reported figure itself. Under the released evaluation, the pooled S5-B and S5-S scores lie 1.75 and 1.33 points below the scores of their respective full-support frames.

13.1. The same frame under both chunk positions

Every estimate above compares different frames and therefore rests on a parallel-trends assumption. Absent the support difference, frames that fall early and late in a chunk are assumed to differ only by a model-independent offset. The placebos make that assumption testable, but they do not remove it.

The release’s chunk starts follow from the index files, so moving them is a one-line change to the protocol and not to the model. Shifting every start 5 windows later sends the frame the release gave position p to position $(p-5) \bmod 21$, which carries the starved block, positions 0–3, onto 16–19. Those frames are then scored twice. The first arm gives one to four windows of recurrent history, as the release scores them, and the second seventeen to twenty. Both use the same weights, the same ground truth and the same evaluator. 3 574 labeled frames satisfy both, and the comparison is within model and within frame. Shifting *earlier* does not work: the release’s own first start is already clamped at zero for most sequences, so most frames would keep the position they had.

Both paired estimates move in the same direction as the chunk-position contrast. Their magnitudes are not compared directly because the velocity-evaluable and all-label populations differ. The remaining two rows of Tab. 5 show

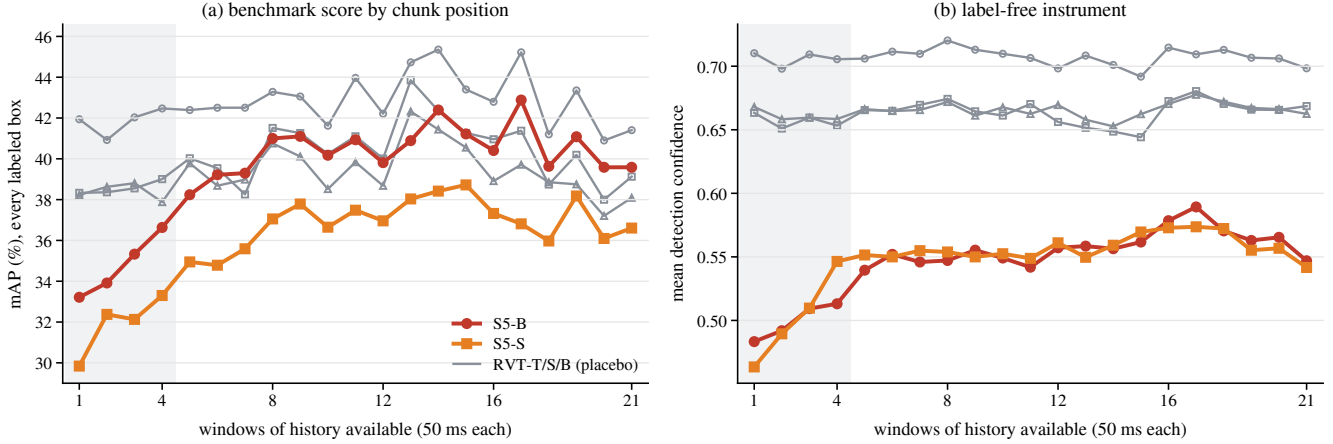


Figure 4. The chunk-position contrast at each of the 21 positions. (a) mAP over every labeled box, the top panel of main Fig. 1 at full width. (b) Mean detection confidence at the same positions, which uses neither the ground truth nor the evaluator. Both panels separate the two SSM checkpoints from the three RVT placebos in the same direction, so the contrast is present upstream of the matching rule. Shaded: the one-to-four-window block.

paired quantity, 3 574 frames	S5-B	S5-S
Δ mAP, velocity-evaluable (pt)	5.23 ± 0.49	4.81 ± 0.52
Δ mAP, every labeled box (pt)	4.20 ± 0.36	4.10 ± 0.32
Δ mean confidence	0.0579 ± 0.0048	0.0509 ± 0.0035
Δ mean confidence, matched	0.0409 ± 0.0040	0.0519 ± 0.0043
Δ detections per frame	-3.49 ± 0.25	-5.09 ± 0.26

Table 5. The identical frames scored under one to four windows of recurrent history and under seventeen to twenty, the sign taken as the long arm minus the short one. Neither a control model nor a parallel-trends assumption enters this paired comparison. Cluster bootstrap over the same 406 sequences with 300 replicates. The mAP row has $z = 10.7$ and 9.3 , mean confidence 12.0 and 14.6 . Matched confidence is over detections paired to a label of the same class at $\text{IoU} \geq 0.5$.

re-runs the release’s own boundary placement through the identical pipeline and serves as the negative control.

windows of history gained	positions	S5-B	S5-S
+0 (control)	0–3	0.00	0.00
+6	6–9	3.73 ± 0.35	3.54 ± 0.39
+11	11–14	4.91 ± 0.50	4.29 ± 0.52
+16	16–19	5.23 ± 0.49	4.81 ± 0.52

Table 6. Δ mAP on velocity-evaluable boxes, in points, for the same 3 574 frames re-scored after the chunk boundary is moved by four amounts. The sign is the long arm minus the short one. Cluster bootstrap over the 406 sequences with 300 replicates; the +16 row is the paired estimate of Tab. 5. The control arm is exactly zero rather than small: with $S = 0$ the rotation is the identity, so every frame is compared with its own score.

what moves. Confidence rises on the detections that match a label, by 0.0409 and 0.0519 , and total output falls by 3.49 and 5.09 boxes per frame. Both terms of precision move the same way. The starved arm emits more boxes and scores the real ones lower. Neither channel is available to a reader of the pooled number, which averages over the chunk position that sets them.

13.2. The same frames under four boundary positions

The comparison above moves the boundary by one amount, so it establishes that the starved block is recoverable without fixing how much history the recovery needs. Because the rotation $(p - S) \bmod 21$ is modular, the amount is a free parameter. At $S \in \{15, 10, 5\}$ the starved block, positions $0-3$, lands on $6-9$, $11-14$ and $16-19$, which is $+6$, $+11$ and $+16$ further windows of recurrent history. The three amounts select the same 3 574 labeled frames, so the rungs are one population under three doses rather than three separate comparisons. `src/e60_dose.py` asserts set equality of the selected frames before recording the curve. $S = 0$

Both checkpoints increase monotonically in the history gained, and the increments fall as the history grows: on S5-B the first $+6$ windows recover 3.73 of the 5.23 points, and the remaining $+10$ windows recover the rest. S5-S behaves the same way at a lower level. Two properties of the measurement follow. The effect is graded in the quantity the intervention manipulates, which a frame-difficulty confound aligned with chunk position would not produce. That confound is fixed once the frames are fixed, and these rungs share their frames. And the response is closer to saturation at $+16$ than at $+6$, so the deficit is concentrated in the first windows after a boundary. The curve is measured at four amounts on one dataset and one architecture family, and the saturation point is therefore not estimated beyond the ordering of these rungs.

13.3. The same intervention on RVT, in two state regimes

Main Sec. 3 differences the observational chunk-position contrast against the RVT checkpoints. That differencing is valid only if an RVT boundary is not a re-

set, which is a property of the released implementation: `modules/detection.py` keeps the validation state in `mode_2_rnn_states` and clears it only on `is_first_sample`. The property is measured here rather than taken from the source.

RVT’s per-window calls are grouped onto the same 21-window grid the SSM release uses, so a chunk position can be recorded for each frame and a boundary shift applied. With no shift and the state carried, the grouping is inert by construction, and `src/e65_regress_check.py` confirms it on the artifacts. Every one of the 101 591 RVT-s and 74 805 RVT-b detections is bit-identical to the stored per-window dump. The ground truth matches, and the recorded positions equal the release placement frame for frame. Without that identity a null below would be indistinguishable from a driver that dropped the treatment. Two regimes are then compared on the 3 574 frames the shift carries from positions 0–3 to 16–19, under the same selection rule and the same estimand as Tab. 6. The first is *carry*, the released behavior. The second is *reset*, with the recurrent state dropped at every chunk start, which places RVT in the condition the SSM release is already in.

state at the boundary	checkpoint	$\Delta\text{mAP (vel.)}$	z	all boxes
carried (released)	RVT-s	-0.01 ± 0.03	-0.3	0.07
carried (released)	RVT-b	0.05 ± 0.04	1.3	0.03
dropped at the boundary	RVT-s	5.39 ± 0.45	12.0	4.84
dropped at the boundary	RVT-b	6.91 ± 0.64	10.9	6.31

Table 7. The chunk-boundary intervention of Sec. 13.2 applied to RVT. The same 3 574 frames are scored twice, once with the boundary at the release’s own placement and once with it 5 windows later. The sign is the long arm minus the short one. Cluster bootstrap over the 406 sequences with 300 replicates. A random-frame permutation null with 200 draws per arm, matched in size to the treated set, is centred within 0.112 points of zero with a spread of at most 0.33 points in every arm.

With the state carried, as the release leaves it, neither checkpoint moves: both intervals cover zero and the larger magnitude is 0.05 points. With the state dropped at the boundary the same weights on the same frames gain at least 5.39 points, a magnitude comparable to the 5.23 and 4.81 points the SSM checkpoints show under the identical intervention. The carried arm therefore supports the parallel-trends control of main Sec. 3 as a measurement, and the reset arm shows that the shift reaches the model. The null is a property of the released evaluation rather than of an intervention that failed to reach the model. Within this protocol the boundary deficit follows whether the recurrent state survives a boundary and not the architecture. The comparison covers two RVT capacities on Gen1 validation at one shift amount, and no claim is made about training-time behavior, which is not what either release’s streaming evaluator exercises.

14. The recurrent influence tail

Sec. 15.3 reports that the window-ablation-sensitivity centroid tracks the horizon instead of converging. Two horizons alone cannot distinguish a slowly converging support from a divergent one, so the tail is characterized directly.

Let $I(L)$ be the summed sensitivity at lag L windows. A weighted least-squares fit of $\log I$ on $\log L$ over lags 1–9 gives $I(L) \propto L^{-a}$ with $a = 0.628$, and the convergent alternative $I(L) \propto e^{-L/\lambda}$ was fitted on the same lags. Neither was fitted on lags 10–19, which are held out. On the held-out lags, the power law gives an RMS standardized residual of 2.5, compared with 23.8 for the exponential, a $9.4\times$ difference. Fitting all lags 1–19 gives $a = 0.645 \pm 0.005$, with a bootstrap interval of $[0.636, 0.655]$ that no replicate carried to $a \geq 1$. The standard error and that interval come from a parametric resample of the per-lag means and so describe the precision of the fit, not independent observations: all lags are computed from the same 576 samples. These lags are not treated as independent observations.

For a support whose first moment has converged, the centroid ratio approaches 1 as the horizon increases, and a particular finite doubling need not give exactly 1. The measured ratio is 1.864 $[1.851, 1.876]$, an interval excluding 1, and follows from two measured centroids without a fit. Over the horizons measured here the centroid is still moving with the horizon instead of settling. Two finite horizons cannot separate a slowly converging tail from a divergent one, and the exponent is fitted over the observed lag range. The measured horizons do not determine a converged full-history centroid. Conditionally, if the observed $I(L) \propto L^{-a}$ with $a < 1$ persisted asymptotically, then both $\sum_L I(L)$ and $\sum_L L I(L)$ would diverge. The centroid over a horizon of H windows of width w would be $-wH(1 - a)/(2 - a)$, linear in H , and the first moment would then not exist. Either way 77.3 % of the summed window-ablation sensitivity already lies outside the current window, which is what the identifiability claim rests on.

A fill artifact could in principle produce persistent long-lag sensitivity, so the control defined in Sec. 2 applies here. The zero and swap fills decay by factors of 10.3 and 2.2 over the same lags, while substantial long-lag sensitivity remains detectable under both fills.

15. Predictor-side temporal descriptors

15.1. Identifiability and the timing estimands

Main Sec. 4 defines the three latent temporal quantities of a benchmark tuple and the term protocol-identifiable. This subsection fixes the notation for the two predictor-side descriptors.

Let a predictor P emit a state at query time t . A signed effective-weight model uses bin centers c_k and weights a_k with $\sum_k a_k = 1$, giving the effective timestamp offset $\tau_P =$

$\sum_k a_k c_k$ under constant velocity. The empirical ablation profile instead assigns non-negative sensitivities $s_k \geq 0$ to bins and has centroid $c_s = \sum_k s_k c_k / \sum_k s_k$. These are distinct quantities: zero-ablation sensitivity depends on both the predictor and the input, and is not in general a decomposition of temporal weights. Sections 15.2–15.3 measure this operational input-sensitivity profile over the newest window. Section 15.4 estimates the regression coefficient τ under Eq. 5. Interpreting $-\tau$ as an effective state-time offset requires the absence of residual velocity-proportional spatial bias. Section 15.5 compares $-c_s$ with the regression coefficient τ , equivalently c_s with the regression-implied timestamp $\tau_P = -\tau$.

15.2. The input window and its indexing instant

Two released files fix the input side. The dataset configuration `config/dataset/gen1.yaml` of the detector’s own repository [1] names the representation `stacked_histogram_dt=50_nbins=10`, a window of 50 ms divided into ten bins of 5 ms. Line 405 of `scripts/genx/preprocess_dataset.py` constructs the array `ev_repr_timestamps_us_end` by counting backwards from a label timestamp in steps of the window duration, and extends it between consecutive label timestamps. Each representation is therefore indexed by the end of its window, and those ends are placed on label times. A window attached to a label at time t holds the events of $[t - 50 \text{ ms}, t]$, whose uniform weighting centroid falls 25 ms before the instant t at which the label asserts a state. We read both statements from the released source, and the 25 ms figure is a reference point instead of a property of the trained network.

15.3. The measured influence profile

The newest-window ablation-sensitivity centroid is computed from the measured bin-ablation sensitivities of a released checkpoint. We take 480 samples of the released Gen1 validation split with the released `rvt-t` checkpoint, warm the recurrent state for eight steps, and ablate each bin in turn. For sample i , let Y_i denote the emitted detection tensor from the unablated input and $Y_i^{(-k)}$ the tensor obtained after zeroing temporal bin k while keeping the incoming recurrent state identical. We define the ablation sensitivity, or influence, of bin k as

$$s_k = \frac{1}{N} \sum_{i=1}^N \frac{\|Y_i^{(-k)} - Y_i\|_2}{\|Y_i\|_2}, \quad N = 480, \quad (4)$$

$$c_s = \frac{\sum_k s_k c_k}{\sum_k s_k},$$

where c_k is the center time of bin k relative to the label instant. The ablated forward pass receives the same incoming recurrent state as the corresponding reference pass, and its

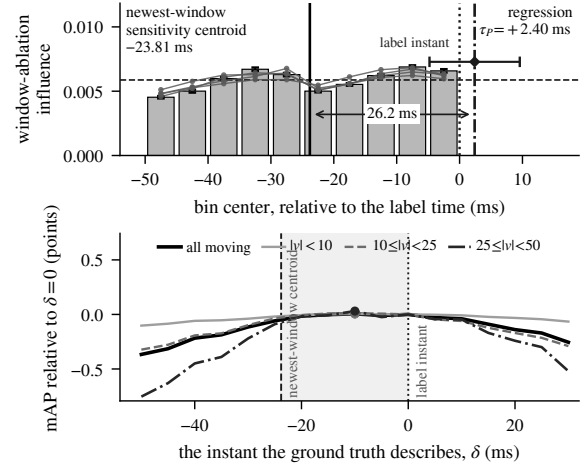


Figure 5. Input sensitivity and the regression-based temporal coefficient summarize the same nominal timestamp without being equivalent, and are not competing estimates of a common latent timestamp; their point estimates are 26.2 ms apart. Panel (b) shows that average precision on the velocity-evaluable subset is nearly invariant over the measured 23.81 ms label-to-centroid displacement. The 26.2 ms separation is not an estimate of architectural latency. (a) Window-ablation influence of each of the ten bins of the released `rvt-t` input window, over 480 Gen1 validation samples with the recurrent state warmed eight steps, with sample-level standard errors. Error bars are descriptive standard errors across the sampled instances and are not used for inference. Uncertainty of the ablation-sensitivity centroid is evaluated by sequence bootstrap in Sec. 2. Dashed: the mean over bins. Dotted: the label instant. Solid: the ablation-sensitivity centroid. Dash-dot, with its standard error shaded: the regression-implied timestamp under Eq. 5, $\tau_P = -\tau$, of Sec. 15.4. Gray: the four other checkpoints, each rescaled to its own mean. (b) Average precision against the temporal center displacement δ applied to the ground-truth boxes, by object speed in px/s, each curve relative to its own value at the label instant, with its maximum error shaded. The shaded band runs from the centroid to the label instant, spanning 0.61 px at the ninetieth percentile of label speed against an annotation dispersion of 0.6353 px (Sec. 3). The mAP span is 0.37 points over all moving boxes and 0.78 points in the 25–50 px/s stratum, where the centroid displacement costs 0.08 points.

returned state is discarded. The profile is a property of the current window and not of a perturbed history.

The profile is balanced between the two halves of the window (Fig. 5a) and yields an ablation-sensitivity centroid of -23.81 ms, compared with -25.00 ms for uniform weighting. The released configuration therefore places the measured centroid near the window reference without implying uniform network sensitivity. Across zero, mean, cross-sample replacement, and gradient instruments, the centroid spans -26.35 to -22.35 ms (Sec. 2). The reported centroid applies to the deterministic 12-sequence subset described in Sec. 2. This is the current window’s window-ablation sensitivity profile with the recurrent state held fixed, not a decomposition of the full temporal support.

Recurrent support beyond the newest window. Over a 1000 ms horizon, the newest window accounts for 22.7% of

the summed window-ablation sensitivity, while the remaining 77.3% lies in preceding windows. The horizon centroid shifts from -161 ms at 500 ms to -299 ms at 1000 ms, so the measured range does not determine a converged full-history centroid. Sec. 14 characterizes the tail and its fill controls.

Across the released checkpoints and a second architecture. RVT releases three Gen1 checkpoints differing only in embedding width, attention head dimension and FPN depth. SSM-ViT [2] releases two more in which each stage’s ConvLSTM is replaced by an S5 state space layer, at 9.68 and 18.19 M parameters against RVT’s 9.87 and 18.54. It distributes RVT’s own preprocessed Gen1 and inherits its window construction. On the same sequences, and with each driven as its own release specifies, all 5 checkpoints put the newest window’s ablation-sensitivity centroid between -23.76 and -24.72 ms. None sits further than 1.24 ms, or 25 % of one bin, from the uniform-weight value.

The support within the newest window is not fixed across capacities. Over 576 samples of the three RVT capacities, the newest window’s share of a 1000 ms horizon falls from 22.7 % to 19.0 %, and the sensitivity centroid moves from -299 to -339 ms.

15.4. Regression-based temporal coefficient

The influence profile is a statement about the network’s input. Even a predictor with uniform local sensitivity can emit a state describing a later instant, and that quantity is measured from the same checkpoint’s own detections.

A detector reporting the state an object occupied τ earlier incurs the physical displacement $-\tau v$, where v is the latent true velocity. We observe only the finite-differenced label velocity $\tilde{v} = v + \eta$ and therefore fit Eq. 5 using $\tilde{v}$. Thus $\tau = -\tau_P$ on the axis of Sec. 15. With $\hat{c}$ the predicted center, c^* the label center, A the area of the ground-truth box, $\hat{u} = \tilde{v}/\|\tilde{v}\|$ and R a quarter turn, b is a two-dimensional intercept. That intercept is fitted globally or as per-sequence intercepts in the reported specification.

$$\hat{c} - c^* = b - \tau \tilde{v} + \rho R \tilde{v} + \gamma \sqrt{A} \hat{u} + \varepsilon \quad (5)$$

It is fitted by stacking both components of every match into one least squares problem, with standard errors clustered by sequence. It is the temporal calibration of [3] carried to a box center, taken as a slope, not as the per-object ratio $-\langle \hat{c} - c^*, \hat{u} \rangle / \|\tilde{v}\|$, whose variance diverges as its denominator approaches zero. For comparison, the per-object ratio evaluated on the same rows has a mean of $+24.22 \pm 27.73$ ms, on the hypothesis evaluated in Sec. 15.5, with an interquartile range of 430 ms. Written this way, Eq. 5 fits the two-dimensional residual directly with $\tilde{v}$ as a regressor. Unlike the per-object ratio, it does not first project the residual onto $\hat{u}$ and divide by $\|\tilde{v}\|$. The reported full specifica-

Specification	τ (ms)	placebo ρ (ms)
<i>velocity estimator, no geometric controls</i>		
forward difference	-14.68 ± 4.99	-6.53 ± 1.68
backward difference	$+45.65 \pm 5.59$	-6.15 ± 1.71
centered difference	$+17.51 \pm 5.06$	-7.76 ± 1.92
<i>controls added, centered difference throughout</i>		
+ box side along travel	$+14.62 \pm 5.47$	-7.75 ± 1.91
+ four geometric terms	-7.62 ± 7.01	-4.12 ± 1.90
+ per-sequence bias	-2.40 ± 7.18	-2.66 ± 1.71

Table 8. The regression-based temporal coefficient τ of the released `rvt-t` checkpoint over 22 534 matched boxes of 376 Gen1 validation sequences, and the placebo ρ beside it, the coefficient on the rotated label velocity, which no temporal offset can reach. Standard errors are clustered by sequence. The forward and backward fits above are calibrated in Sec. 4.

tion adds four coordinate/size regressors and per-sequence intercepts to the terms displayed in Eq. 5. The coefficient is a temporal offset only if no velocity-proportional spatial bias remains after these controls. Such a bias is indistinguishable from τ in Eq. 5.

$R\tilde{v}$ is the label velocity rotated a quarter turn. Because a pure temporal offset acts along $\tilde{v}$, not $R\tilde{v}$, ρ serves as a negative-control coefficient. A nonzero value implies residual spatial misspecification. $\sqrt{A} \hat{u}$ absorbs a displacement along travel proportional to box side, which is needed because that residual grows with box size and box size correlates with label speed at $+0.139$ on this split.

The finite difference used for the velocity. $\tilde{v}$ is finite differenced from labels. Label noise enters forward and backward velocity differences with opposite shared-noise signs. The centered difference excludes the current label center and removes this direct same-frame term. Secs. 1 and 4 derive and calibrate this effect, so the reported regression uses the centered estimator.

Regression dataset and matching. Rows are detections matched to nominal-time ground truth at $\text{IoU} \geq 0.3$ with centered label velocities available, leaving 22 534 matches from 376 sequences. The estimate is conditional on this matched subset. Sec. 5 reports association-rule sweeps, including overlap-free matching.

Table 8 is a specification ladder, and the specification was chosen by the placebo, not by τ . Under Eq. 5 with only the box-side term, $\tau = +14.62 \pm 5.47$ ms, but ρ is -7.75 ms at 4.0 standard errors, which indicates a term omitted from Eq. 5 that $R\tilde{v}$ is a proxy for. A split by the sign of the horizontal velocity locates it: ρ changes sign across that split, at -27.99 ms and $+21.84$ ms, which a rotational term would not do. Most label motion here is horizontal, so $R\tilde{v}$ maps a horizontal speed onto a vertical residual. The rotated regressor represents a bias that differs between oncoming and same-direction traffic. Entering box height and im-

age row along $\hat{y}$ and box width and image column along $\hat{x}$, and then a constant vector per sequence, returns the placebo to -2.66 ms at 1.6 standard errors. That is the specification reported, and the criterion that selected it is a coefficient on a regressor orthogonal to $\tilde{v}$.

Under the reported specification, $\tau = -2.40 \pm 7.18$ ms, with a sequence-bootstrap 95% interval of $[-17.0, +11.5]$ ms. Across geometric-control specifications, τ spans -20.2 to $+11.1$ ms. Because the reported specification was selected using the orthogonal placebo coefficient, its nominal standard error does not include specification-selection uncertainty. We treat the point estimate descriptively and use the full specification sweep to assess robustness.

15.5. Relating input-sensitivity timing to the regression-based temporal coefficient

The comparison combines measurements under different recurrent-state conditions. The centroid uses an eight-step warm-up with the incoming state fixed during ablation. The regression uses detections from the validation stream with the released state carried across frames. The 26.2 ms numerical separation is descriptive only: the two quantities are distinct operational summaries and are not competing estimates of a common latent timestamp. Under the reported specification and its annotation-error assumptions, the point-estimate difference is 26.2 ms. The regression coefficient has a standard error of 7.2 ms. This is a regression-coefficient result instead of a latency estimate of this architecture. Expressed on Eq. 5's sign convention, the newest-window centroid corresponds numerically to $\tau = +23.81$ ms. This comparison relates two operational estimands and does not assume that Eq. 5 must recover the ablation centroid. Thus, 23.81 ms is the measured interval between the label instant and the newest-window ablation-sensitivity centroid, whereas 26.2 ms is the point-estimate difference between that centroid and the reported regression coefficient. The measured horizons do not determine a converged full-history centroid. The variation across specifications is substantial relative to the reported standard error and is not captured by that standard error.

The detector is trained against the ground truth at the label time on a window ending there, so a measured offset near zero is consistent with the training objective indexed at the label time. The objective does not enforce the interval, and the instant the output describes and the centroid of the evidence used to produce it are different quantities that one benchmark timestamp stands for. At the ninetieth percentile of label speed on this split, 25.7 px/s, the 23.81 ms is 0.61 px of displacement, comparable to the annotation dispersion of the labels themselves (0.6353 px), and at the ninety-ninth it is 1.29 px. That ablation-sensitivity centroid and the regression-based temporal coefficient are measured

from released artifacts alone.

Effect on benchmark mAP. This experiment is not a test of whether temporal support matters in general. It asks whether Gen1's particular speed composition makes the measured RVT interval visible to its current mAP. We translate each ground-truth box center by $\delta\tilde{v}$, keeping its size and class fixed, a first-order surrogate for temporal displacement under local constant velocity. The released predictions remain unchanged, since a detection has no velocity of its own. Over 20 296 frames, 68.7 % of the 40 698 ground-truth boxes carry a centered velocity. Ground-truth boxes without a centered velocity are marked ignored during evaluation, and detections matched to ignored boxes are excluded from TP/FP accounting. The resulting mAP against δ is smooth with its maximum at -10 ms (Fig. 5b). Along the axis of Eq. 5 that is $\tau = +10$ ms, and mAP varies by at most 0.04 points over the interval. This descriptive curve does not by itself identify a preferred temporal value.

The full -60 to $+60$ ms sweep moves mAP by 0.9 points. On the -50 to $+30$ ms grid summarized in Table 9, the all-moving span is 0.37 points. Displacing the ground truth to the window's centroid costs 0.0004 absolute mAP (0.04 percentage points, 0.12 % relative to 0.3346). Average precision on the velocity-evaluable subset is nearly invariant over the measured label-to-centroid displacement, and at the common -23.81 ms displacement the ordering of all 5 released checkpoints is unchanged on the validation split. Scoring each checkpoint at its own preferred displacement likewise leaves the validation-split ordering unchanged in the reported strata. Both statements are bounded by the stability of the ordering under sequence resampling at this scale. The smallest timing difference that would reorder an adjacent pair is 43.7 ms over all moving boxes, $1.83\times$ the measured interval, and 11.2 ms in the 25–50 px/s stratum. Separately, on every labeled box instead of the velocity-evaluable subset the displacement acts on, the benchmark's own cluster resampling holds the validation-split ordering in only 42.5 % of replicates. Its tightest adjacent pair is separated by 0.050 points against a paired standard error of 0.53. A common displacement sweep reveals one localized inversion between rvt-b and S5-S in the 25–50 px/s stratum. Table 9 locates the flatness in that composition. Of the moving boxes, 72.1 % travel below 10 px/s, where the displacement is 0.05 px. The sweep deepens to 0.78 points between 25 and 50 px/s, where it is 0.77 px, past the labels' own annotation dispersion of 0.6353 px (Sec. 3). The near invariance therefore characterizes Gen1's metric resolution, not the absence of a temporal discrepancy.

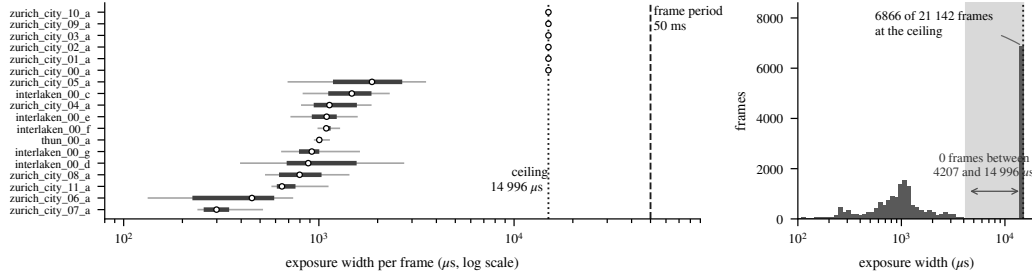


Figure 6. Published per-frame exposure width across the eighteen DSEC training sequences whose exposure files resolve: per sequence (left, log axis, median, interquartile range and 5–95 % band, against the fixed 50 ms frame period) and pooled over 21 142 frames (right). 32.5 % of frames sit at the auto-exposure ceiling of 14 996 μ s and zero fall between the largest non-ceiling width of 4207 μ s and that ceiling, separating the two exposure regimes. The six ceiling sequences are those measured in Sec. 16.

speed (px/s)	share (%)	med. speed	centroid (px)	Δ mAP	span
all moving	100.0	4.0	0.10	0.04	0.37
< 10	72.1	2.1	0.05	0.02	0.11
10–25	19.2	15.0	0.36	0.01	0.33
25–50	7.5	32.3	0.77	0.08	0.78
≥ 50	1.2	57.4	1.37	0.00	0.37

Table 9. The sweep of Fig. 5b by object speed over the 27 943 ground-truth boxes with centered velocity. Other boxes are ignored. Δ mAP is the cost of shifting the ground truth to the newest-window ablation-sensitivity centroid at -23.81 ms. Span is the mAP range over the -50 to $+30$ ms sweep. Both values are reported in points.

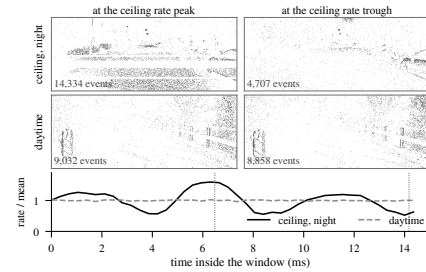


Figure 7. Top: 350 μ s slices of the ceiling-exposure sequence zurich.city_09_a and of the daytime control, at the peak and trough of the ceiling sequence's own event rate inside its densest published exposure window. Bottom: that rate in 350 μ s bins, each sequence normalized to its own mean. The ceiling rate moves by a factor of 3.0 between its extremes, the daytime rate by 1.1.

16. Label-acquisition timing within one exposure

16.1. Exposure and labels

The predictor-side analysis concerns the temporal support of released detector inputs and outputs. DSEC provides a separate exposure-side analysis. Published exposure intervals bound the interval within which event timing is observed. Within-exposure event timing contains illumination-locked structure and therefore cannot be uniquely attributed to object timing without separating that structure. DSEC [4] publishes per-frame exposure metadata, and the files resolve for eighteen train sequences. The width varies by a factor of 127 against a frame period fixed at 50 ms, and within one recording by up to a factor of 12.5. The auto-exposure ceiling of 14 996 μ s holds 32.5 % of frames, in six of the eighteen sequences, and the distribution is not continuous over that range (Fig. 6). The gap separates the two exposure regimes, and ceiling exposure forms a distinct stratum.

The released DSEC-Det labels comprise 390 118 boxes on 6693 tracks across 60 sequences. Every label sits on the frame clock, and the median spacing between consecutive label times is 50 001 μ s. Over 361 628 triples of consecutive labels on a track, 7.1 % of middle box centers fall within a hundredth of a pixel of their neighbors' linear interpolant.

This rules out exact local linear interpolation as a general description of the released trajectories without identifying the underlying annotation process. DSEC-Det ships more than one annotation version [5], Sec. 8 names the one measured here. Median box-center image speed is 40 px/s, with 360 px/s at the ninety-ninth percentile.

16.2. Mains-frequency signature and the per-pixel phase test

Swiss mains is 50 Hz, and rectified lighting can produce the tested 100 Hz modulation. The pre-specified 100-Hz line and its harmonics appear in the event-rate spectrum of every ceiling sequence, while the daytime control lacks the same pattern (Fig. 7). Per-pixel phase tests and the pre-specified frequency control establish the same illumination signature.

Per pixel, the Rayleigh statistic $Z = R^2/N$ is $\text{Exp}(1)$ under independent uniform phases. The off-frequency and daytime controls empirically calibrate this nominal null, while the ceiling sequence has median $Z = 2.935$ at 100 Hz and 24.07 % of live pixels reach the nominal $p < 10^{-3}$ threshold under the independent-uniform-phase reference. The illumination attribution therefore relies primarily on the

frequency-matched and daytime controls, not on the nominal Rayleigh reference alone.

16.3. All six ceiling sequences, per labeled box

The same statistic taken per labeled box, over the events inside that box during that frame’s own exposure window, extends to every ceiling sequence. Across the six sequences and 16 516 boxes, 95.4–100.0% satisfy the pre-specified criterion in Table 2.

16.4. Within-box temporal dispersion under that signature

That signature limits attribution of within-exposure event timing to object timing. The per-box 137-Hz arm is used as a same-event descriptive control, not as an independent spectral-resolution test. The illumination attribution is supported primarily by the long-slice 100-Hz harmonic structure and the daytime control. Within-box dispersion is $206.6\ \mu\text{s}$ against a null of $93.0\ \mu\text{s}$, with an excess of $183.6\ \mu\text{s}$. A frame-level permutation preserving pooled temporal distributions and box counts gives a null of $78.6\ \mu\text{s}$, and the excess remains after the clean-pixel control. Random-position boxes show that the dispersion is not specific to annotated object regions (Sec. 9), which supports reading the signature as an attribution limit and not as object motion.

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